



Continuous gas and particle monitoring in the metro of Valparaíso

CLIENT

EFE Trenes de Chile

SECTOR

Transport

LOCATION

Valparaíso, Chile

PRODUCTS

Kunak AIR Pro
Kunak AIR Cloud

Need

Empresa de los Ferrocarriles del Estado, known by its acronym EFE and called EFE Trenes de Chile, needed to conduct a study on the air quality in the Valparaíso metro stations.

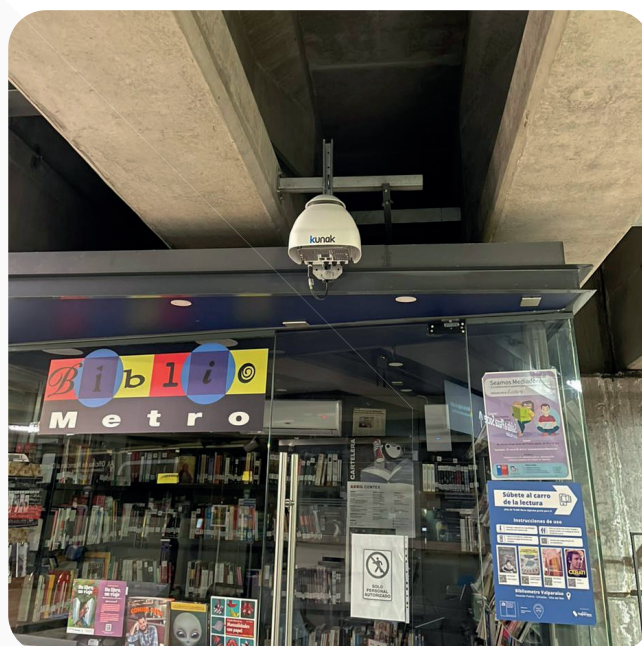
The Valparaíso metro currently has 2 main uses:

- Passenger transport during the day with an electrified system.
- Transport of materials during the night with a diesel engine.

The objective of the study is to improve the ventilation system installed on the metro lines in order to improve the air quality and comply with the Chilean law 594 to protect the health of both workers and users of the metro.

Measured parameters

- CO (carbon monoxide)
- CO₂ (carbon dioxide)
- NO₂ (nitrogen dioxide)
- Particulate Matter (PM₁, PM_{2.5}, PM₄, PM₁₀, TSP and TPC).
- Temperature, relative humidity, dew point and atmospheric pressure.





The aim is to enhance the ventilation system implemented on the metro lines, with the purpose of improving the air quality and ensuring compliance with Chilean law 594.



Solution

This study of air quality will be carried out on the basis of 3 lines of work:

- Fluid-dynamic simulation and modelling of ventilation systems in underground stations.
- Measurement of exposure to pollution of metro staff (occupational hygiene) by means of certified personal exposure devices.
- Monitoring of gases and particles both in the environment and in the stations themselves.

For this purpose, 4 Kunak AIR Pro stations will be deployed to monitor different gases (CO, CO₂ and NO₂) and particles (PM₁, PM_{2.5}, PM₄, PM₁₀, Total Suspended Particles (TSP) and Total Particles Count (TPC) both outdoors and inside the metro stations.

Benefits

- Creation of a healthier environment by monitoring gases and particulates for a safer atmosphere.
- Optimisation of the ventilation system by adjusting airflow based on real-time data to improve air quality.
- Improvement of preventive maintenance by identifying system problems in advance.
- Compliance with air quality standards to ensure public safety.
- Generation of greater confidence and enhancement of the company's image by demonstrating its commitment to passenger welfare.

Results

The 4 monitoring stations will collect key information with a higher spatial resolution of the pollution and will complement the data collected by the personal exposure devices.

With all the information it will be possible to carry out modelling of the ventilation systems and consequently improvements of the systems can be carried out.

